

ITC MP

Correction DS3

1)

```
SELECT idC, val / taille as rentabilite FROM Conteneurs WHERE  
portDepC = "Marseille" AND portDestC = "Barcelone" AND dateDisp < 2025-01-01  
ORDER BY rentabilite DESC;
```

2)

```
SELECT idN, COUNT() as nbC FROM Conteneurs GROUP BY idN WHERE idN != 0;
```

3)

```
def profit(obj, S):  
    total = 0  
    for i in range(len(obj)):  
        total += obj[i][1] * S[i]  
    return total
```

4)

```
def contrainte(obj, S, b):  
    capacite = 0  
    for i in range(len(obj)):  
        capacite += obj[i][0] * S[i]  
    return capacite <= b
```

5)

```
1 0 1  
1 1 0
```

6)

2^n feuilles, donc $O(n2^n)$

7)

```
1 1 0
```

8)

```
Lqi.append(obj[i][1] / obj[i][0])  
...  
Li[j] = Li[j-1]  
...  
Li[j] = i
```

9)

$O(n)$ ($O(n^2)$)

10)

```
while j < len(obj):  
    if obj[Li[j]][0] <= b:  
        S[Li[j]] = 1
```

```

        b = b - obj[Li[j]][0]
    j += 1

```

11)

12)

13)

$O(bn)$

14)

15)

```

def est_feuille(a):
    return a['g'] == {}

```

16)

```

def possible(obj, Sk, b):
    n = len(obj)
    k = len(Sk)
    Smin = Sk + (n-k) * [0]
    Smax = Sk + (n-k) * [1]
    return contrainte(obj, Smin, b) and profit(obj, Smax) > Pmin

```

17)

```

else:
    if possible(obj, arbre['g']['S'], b):
        KPPse(arbre['g'], obj, b)
    if possible(obj, arbre['d']['S'], b):
        KPPse(arbre['d'], obj, b)

```

18)

```

for i in range(len(T)):
    T[i] *= rho

```

19)

```

kmax = 0
Pmax = 0
for i in range(len(S)):
    P = profit(obj, S[i])
    if P > Pmax:
        kmax = i
        Pmax = P

```

20)

```

for i in range(len(T)):
    if S[kmax][i] == 1:
        T[i] += 1/(1+PbestOfAll-Pmax)
        T[i] = max(T[i], Tmin)
        T[i] = min(T[i], Tmax)

```

21)

```
o0 = randint(0, n-1)
...
S[k][o0] = 1
b2 -= obj[o0][0]
```

22)

```
prob = {}
...
oi = candidats[i]
prob[oi] = (T[oi]**alpha)*((b * obj[oi][1] / obj[oi][0])**beta)
...
oi = candidats[i]
prob[oi] /= s
```

23)

```
prob = construitProb(obj, candidats, b, T)
c = choixCandidats(candidats, prob)
S[k][candidats[c]] = 1
b2 -= obj[candidats[c]][0]
candidats_2 = []
for cc in candidats:
    if cc != c and b2 > obj[cc][0]:
        candidats_2.append(cc)
candidats = candidats_2
```